CHEHAK AGGARWAL

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EDUCATION

University of California San Diego, Jacobs School of Engineering *Expected June 2029*

B.S. Aerospace Engineering · GPA 3.9 · Triton Scholarship (Merit)

Coursework: Linear Algebra, Multivariable Calculus, MAE 2 design labs, MATLAB

ENGINEERING PROJECTS

Yonder Dynamics: Mars Rover Team (URC) — *Mechanical Team Member* *Nov 2025 – Present*

- Designed a four-bar-linkage science claw with dual-servo actuation, replaceable fingers and a bolt-on interface to the rover hand; it collected geological samples at the University Rover Challenge 2026.
- Redesigned the forearm electronics enclosure to fit a new ODrive motor controller and fuse board, lengthening it 21% (178 → 216 mm) and reworking the cable exit and cap for bend radius and stacked wiring; printed a PLA prototype.
- Ran static FEA (von Mises) on the arm's elbow joint across two load cases to locate stress concentrations at mounting interfaces; adapted a lead-screw, four-bar-linkage gripper in Onshape as a team design task.

Yonder Deep: Arctic Autonomous Surface Vessel — *Mechanical Team Member* *Nov 2025 – Present*

- Co-designed the modular welded-and-bolted 6061-T6 aluminum frame joining the vessel's pontoons, carrying sensors, electronics and thrusters at a ~5 cm pontoon draft.
- Supported two pool tests of an X-configuration thruster rig, verifying buoyancy, power draw and hydrophone performance; the rig completed an out-and-back run.
- Designed the LiDAR mount in Fusion 360; contributed to trade studies on enclosure selection, thruster orientation, mass distribution and flip recovery.

JELIX: Jellyfish-Inspired Underwater Propulsion — *Student Researcher, Genex Space (remote)* *Jun 2024 – Jul 2025*

- Designed a propeller-free UUV propulsion mechanism: a four-rib cam–follower drive with spring return that mimics jellyfish pulsing.
- Designed the cam profile in CATIA V5 and simulated it in MSC ADAMS at 5 rpm (65 × 80 mm rib-tip travel, 0–40° stroke, 12 s cycle); built a free-body model relating rib angle to thrust force.

EXPERIENCE

Big Data Rhino — *Intern* *Jun 2026 – Sep 2026*

- Analyzed satellite data in Python to evaluate proof-of-concept products; wrote problem statements, workflow diagrams and technical documentation in GitLab.
- Taught AI classes to 48 participants.

LEADERSHIP

AIAA at UC San Diego — *Project Coordinator (EDI Coordinator, Jan–Jun 2026)* *Jun 2026 – Present*

- Built and run four student teams entering NASA competitions (HuLC, RASC-AL, Gateways to Blue Skies, ORBITS): reviewed 57+ applications, placed 32 students, and assign and track tasks.
- Co-researched and co-wrote a NASA Human Lander Challenge 2026 proposal on tunable viscoelastic damping for life-support noise.

SKILLS

CAD: Onshape, Fusion 360, CATIA V5

Analysis: FEA, MSC ADAMS multibody dynamics, kinematics

Fabrication: 3D printing (FDM), laser cutting, manual lathe

Programming: Python, MATLAB, GitLab

Languages: English, Hindi

AWARDS & INTERESTS

International Aerospace Olympiad, 2nd rank (2024) · Competitive boxer: South Asian Games gold (2022), national gold · Digital artist (64k Instagram followers)